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Reflection Journal: VGG16 Image Classification Lab

This lab was a good experience for me because I was able to work with a deep learning model without having to create the model myself. I used VGG16 with ImageNet and learned how it can recognize objects in an image.

Something that I learned was that the image needs to be prepared before the model can use it. The image has to be resized and converted into the correct format. I also learned how the model gives different predictions with confidence percentages. I used a dog image for my test, and the model predicted Golden Retriever with 88.26% confidence. After I changed the image by shifting the pixels, the prediction was still Golden Retriever, but the confidence changed to 86.04% and 86.44%. I thought this was interesting because the change was small, but the results were not exactly the same.

Another trouble that I had during the lab. The first image I tried to download did not work because of a 429 error. I solved this by uploading my own image. I also had an error with the `random_shift` function because the arguments were different in my Keras version. I checked the function and changed the code so it could run.

In overall, this lab helped me understand better how pre-trained models work and how sensitive they can be to changes in the input image. It also gave me some experience troubleshooting errors instead of just running the code and looking at the results.